Tanish Singh Rajpal

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Reliable quantum information systems: learning, verification, and simulation

EDUCATION

Carnegie Mellon University — *Pittsburgh, PA*
M.S. in Information Technology - Information Security — *Expected May 2027*
Information Networking Institute

NMIMS University — *Mumbai, India*
B.Tech. in Computer Engineering — *May 2025*
GPA: 3.83/4.00 | Valedictorian, graduating class

RESEARCH EXPERIENCE

Certifying delivered entanglement in quantum networks — *May 2026 - present*
Independent Study, CMU Information Networking Institute | Advisor: Prof. Michael Mattarock

- Developing a two-layer method: a model-based guard for individual deliveries and hidden random tests that certify average fidelity of kept pairs without trusting the delivery model.
- In pre-registered simulations on grids up to 30 x 30 nodes, none of 4.2 million pairs delivered after an alarm fell below the fidelity threshold; 74 of 83 primary hypotheses passed, with failures retained in the public record.
- Re-analyzed public experimental data and ran two test-or-keep experiments on one IBM quantum processor. These are component checks, not a physical-network demonstration. Manuscript in preparation; code and evidence.

Differentiable quantum-chemical models (DFTBML3) — *2026 - present*
Independent Study, CMU Chemistry | Advisor: Prof. David Yaron

- Working on differentiable density-functional tight-binding models for chemical Hamiltonians, preserving physical structure while testing trainable components against established implementations. Results in progress.

A hybrid architecture for post-quantum EAP-TLS — *Aug 2026 - present*
Master's Thesis, CMU | Advisors: Prof. Pedro Bustamante and Prof. Michael Mattarock; Reader: Prof. Patrick Tague

- Investigating whether oversized ML-DSA certificates can cause Path MTU Discovery blackholes in WPA-Enterprise 802.1X authentication.
- Designing an ns-3 evaluation of an edge QUIC proxy and QKD-secured wired backbone, with hypotheses covering handshake latency, RADIUS compute, IPv4/IPv6 behavior, and key-delivery capacity. Evaluation in progress.

SELECTED QUANTUM PROJECTS

Phase Hunter - mapping quantum spin phases under noise — *Sep 2026*
Q-SITE Hacks 2026, Quantum Coalition Scientific Track runner-up | Team Popeyes

- With Paul Jeon and Sumedhaa Ruhil, built a playable phase-boundary search backed by reproducible PennyLane simulations. Compared clean, noisy, and error-mitigated phase maps under limited measurement budgets. Code and notebook.

Quantum portfolio optimization using QAOA — *Apr 2026*
Team Lead, Yale YQuantum Hackathon | Third place

- Led the Markowitz-to-QUBO-to-Ising formulation and QAOA circuit design for an insurance portfolio optimization prototype. Repository.

PEER-REVIEWED PUBLICATIONS

Journal article

Rajpal, T. S., and Naithani, A. (2026). "NeuroCrypt: A Neuro Symbolic AI Ecosystem for Advanced Cryptographic Data Security and Transmission." IEEE Transactions on Artificial Intelligence, 7(1), 512-521. doi:10.1109/TAI.2025.3577605

Conference papers

Naithani, A., and Rajpal, T. S. (2025). "Blockchain-Based Digital Health Passport: A Decentralized Framework for Secure and Interoperable Health Data Management." IEEE ZINC 2025. doi:10.1109/ZINC65316.2025.11100281

Naithani, A., Rajpal, T. S., and Mistry, V. (2024). "TAV-Cart: Transaction Assistance and Validation Cart System for Effortless Shopping and Checkout." IEEE ICACCM 2024. doi:10.1109/ICACCM61117.2024.11059075

Naithani, A., Jain, V., Rajpal, T. S., and Mistry, V. (2024). "AI Based Goalkeeper for Penalty Shot Prediction Using RNN and Pattern Matching Algorithm." IEEE ISDFS 2024. doi:10.1109/ISDFS60797.2024.10527307

HONORS AND TRAINING

- Q-SITE Hacks 2026: runner-up, Quantum Coalition Open Challenge (Scientific Track); third place, IBM Quantum Coding Challenge; third place, Overall Coding Challenge; fourth place, IonQ Championship.
- IBM Qiskit Global Summer School 2026 - Quantum Excellence (Advanced). Credential.
- Third place, Yale YQuantum Hackathon (2026); first place, Crypto Quest Cybersecurity Competition (2024).

RESEARCH METHODS AND TOOLS

Methods: quantum-network simulation, pre-registered experiments, model checking, statistical certification, reinforcement learning, PPO, QAOA, QUBO/Ising formulations.

Platforms: NetSquid, ns-3, PennyLane, Qiskit, Python, Git/GitHub.

Security and networking: EAP-TLS, ML-DSA, 802.1X, PMTUD, QKD, Burp Suite, Metasploit.

INDUSTRY EXPERIENCE

Vulnerability Assessment and Penetration Testing Intern *Oct 2023 - Jan 2024*

Newton's Apple, Pune, India

- Identified and supported remediation of three critical vulnerabilities in a confidential Android application; performed static Java code analysis with the development team.

ACADEMIC SERVICE

Student Ambassador, CMU Information Networking Institute *Oct 2025 - present*

Support prospective students and help plan community events.